

Harmonic oscillator 2

From the previous section

$$\psi_n = C_n \exp\left(\frac{m\omega x^2}{2\hbar}\right) \frac{d^n}{dx^n} \exp\left(-\frac{m\omega x^2}{\hbar}\right)$$

where C_n is the normalization constant

$$C_n = \frac{(-1)^n}{\sqrt{2^n n!}} \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \left(\frac{\hbar}{m\omega}\right)^{n/2}$$

Evaluate the derivative of ψ_n . By the product rule

$$\frac{d}{dx}\psi_n = \frac{m\omega x}{\hbar}\psi_n + C_n \exp\left(\frac{m\omega x^2}{2\hbar}\right) \frac{d^{n+1}}{dx^{n+1}} \exp\left(-\frac{m\omega x^2}{\hbar}\right)$$

Substitute ψ_{n+1} for the right-hand side.

$$\frac{d}{dx}\psi_n = \frac{m\omega x}{\hbar}\psi_n + \frac{C_n}{C_{n+1}}\psi_{n+1}$$

We have

$$\frac{C_n}{C_{n+1}} = -\sqrt{\frac{2m\omega}{\hbar}}\sqrt{n+1}$$

Hence

$$\frac{d}{dx}\psi_n = \frac{m\omega x}{\hbar}\psi_n - \sqrt{\frac{2m\omega}{\hbar}}\sqrt{n+1}\psi_{n+1}$$

Rearrange as

$$\sqrt{\frac{\hbar}{2m\omega}}\left(\frac{m\omega x}{\hbar} - \frac{d}{dx}\right)\psi_n = \sqrt{n+1}\psi_{n+1}$$

Let a_+ be the operator

$$a_+ = \sqrt{\frac{\hbar}{2m\omega}}\left(\frac{m\omega x}{\hbar} - \frac{d}{dx}\right)$$

Hence a_+ is the step-up operator

$$a_+\psi_n = \sqrt{n+1}\psi_{n+1}$$

By the commutation relation $[a_-, a_+] = 1$ we have the step-down operator

$$a_- = \sqrt{\frac{\hbar}{2m\omega}}\left(\frac{m\omega x}{\hbar} + \frac{d}{dx}\right)$$

such that

$$a_-\psi_n = \sqrt{n}\psi_{n-1}$$

Let N be the number operator

$$N = a_+a_-$$

Hence

$$N\psi_n = a_+a_-\psi_n = a_+(\sqrt{n}\psi_{n-1}) = n\psi_n$$

Eigenmath script